

AI協作與倫理

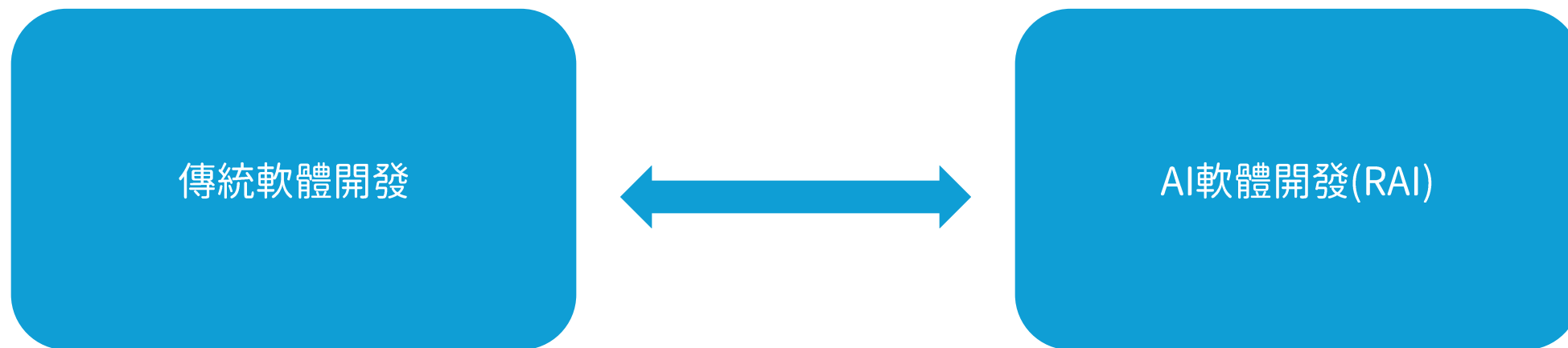
An Empirical Study on Decision-Making Aspects in Responsible
Software Engineering for AI

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大綱

- 研究背景&動機
- 研究方法
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- 總結

研究背景&動機



研究背景&動機

- 倫理準則
 - Ex: 歐盟AI法案
- 日常AI開發實踐
 - 如何實踐?

研究方法

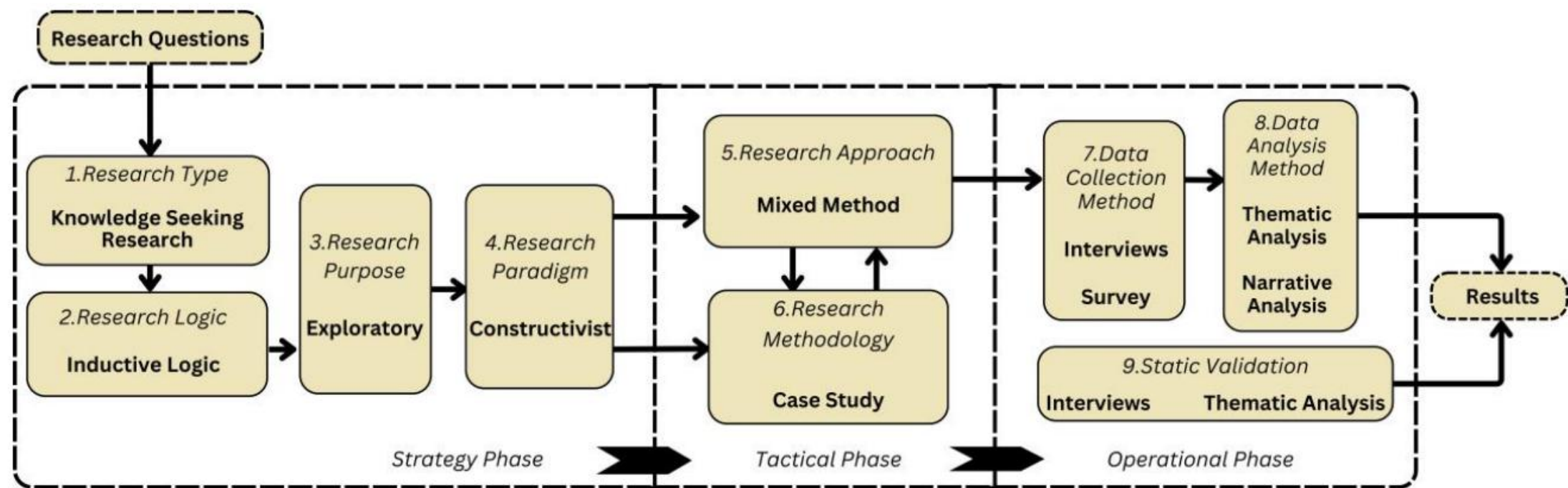


Figure 1. Decision-making structure for the research design

研究方法

TABLE I
INTERVIEW SUBJECT CHARACTERISTICS

Code	Gender	Age	Years in current company	Years in current role	Job Role/Designation	Primary area of expertise	Total exp (years)	Current company size
P1	Male	37	11	1	Head of AI Development	Software	12	200
P2	Male	32	2	2	Data Engineering Consultant	Software & AI	6	10
P3	Male	28	2	5	Full Stack Software Engineer	Software	10	9K
P4	Female	39	4	1	Senior Technical Specialist	Software	11	2K
P5	Male	37	2	2	Full Stack Engineer	Software	13	10K
P6	Female	51	2	2	Senior Product Manager	Software & AI	25	10K
P7	Male	34	5	5	Machine Learning Engineer	Software & AI	8	10

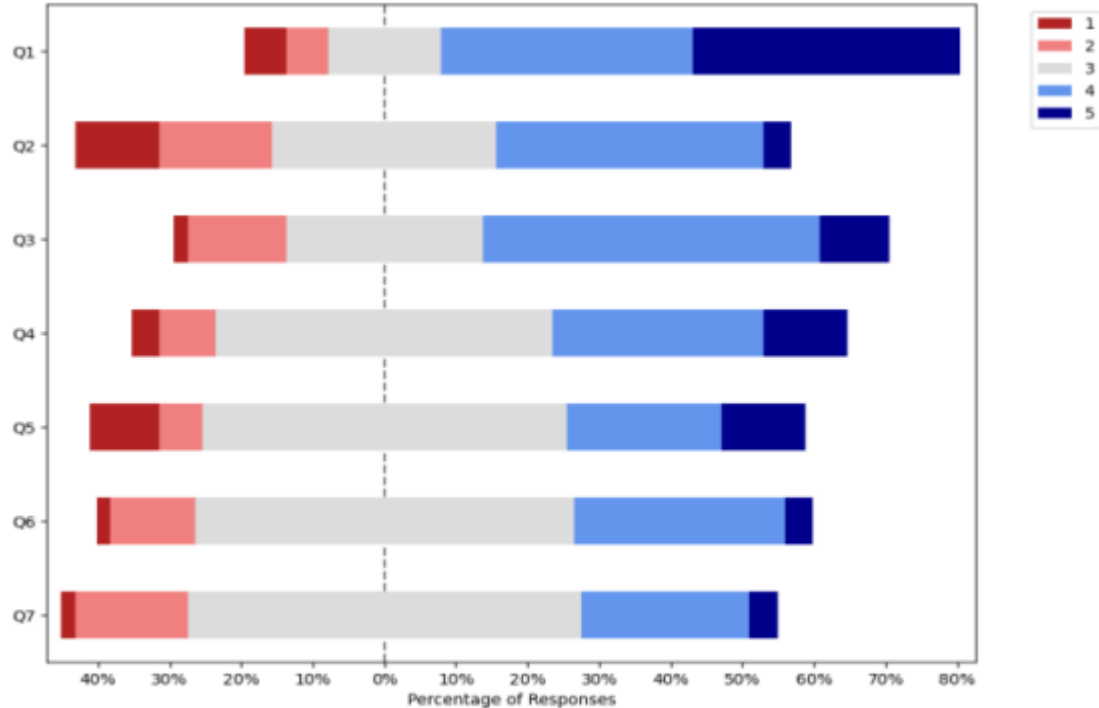
TABLE II
INTERVIEW SUBJECT CHARACTERISTICS FROM STATIC VALIDATION INTERVIEWS

Code	Age	Gender	Current job role	Primary area of expertise	Total experience	Current company (Size)	Years in current role	Years in current role	Current sector of work
SV1	37	Woman	Data Scientist	Software & AI	12+ years	19000	5	5	Tech
SV2	36	Woman	Data Scientist	Software & AI	2.5 years	20000+	1	1	Financial
SV3	47	Man	CTO	Software & AI	16 years	7	2.5	2.5	Healthcare
SV4	28	Woman	Specialist(SE)	Software & AI	6.4 years	3000	1	1	Tech

結果

- RQ1 - Unique aspects of RAI
- RQ2 - Role variation in RAI
- RQ3 - Human factors in RAI

結果 – RQ1



- Q1 How responsible do you feel for the ethical implications of the AI/software you develop?? (1: Not at all - 5: Completely)
- Q2 How much do conflicting personal values within the team affect decision-making for RAI? (1: Negligible impact - 5: Substantial impact)
- Q3 How prepared is your organization to integrate RAI principles? (1: Not Prepared at all - 5: Very Well Prepared)
- Q4 How would you rate your understanding of ethical software/AI principles? (1: Very poor - 5: Excellent)
- Q5 Are you aware of any AI regulations in your region, and how do they affect your work? (1: Significantly hinders - 5: Significantly facilitates)
- Q6 How do you rate the decision-making process in the development of RAI in your organization? (1: Very poor - 5: Excellent)
- Q7 How effective are the methodologies or frameworks your organization uses to ensure RAI development? (1: Not effective - 5: Very effective)

Figure 2. Diverging stacked bar chart - Likert questions for RQ1

結果 - RQ2

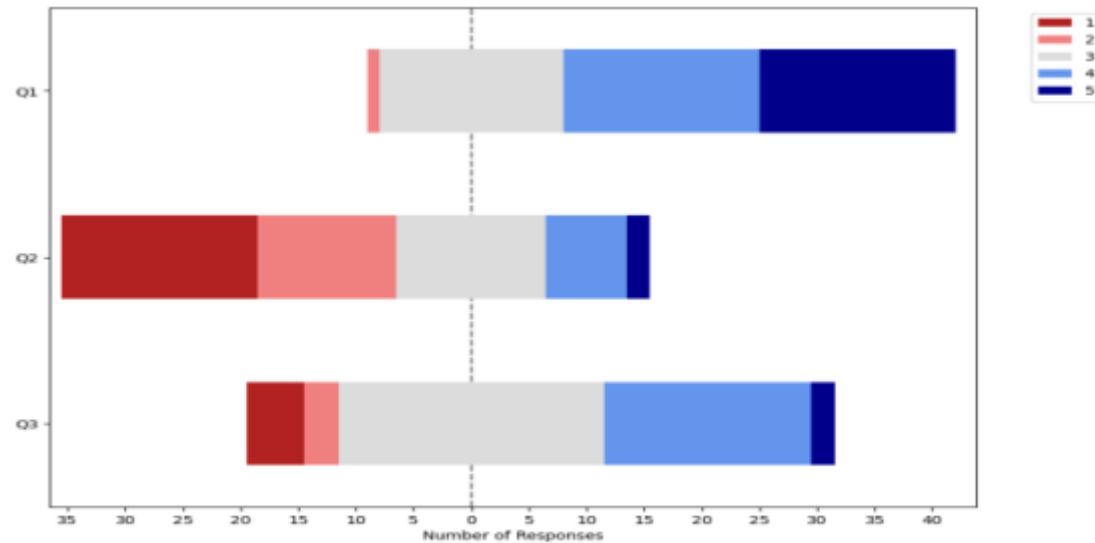
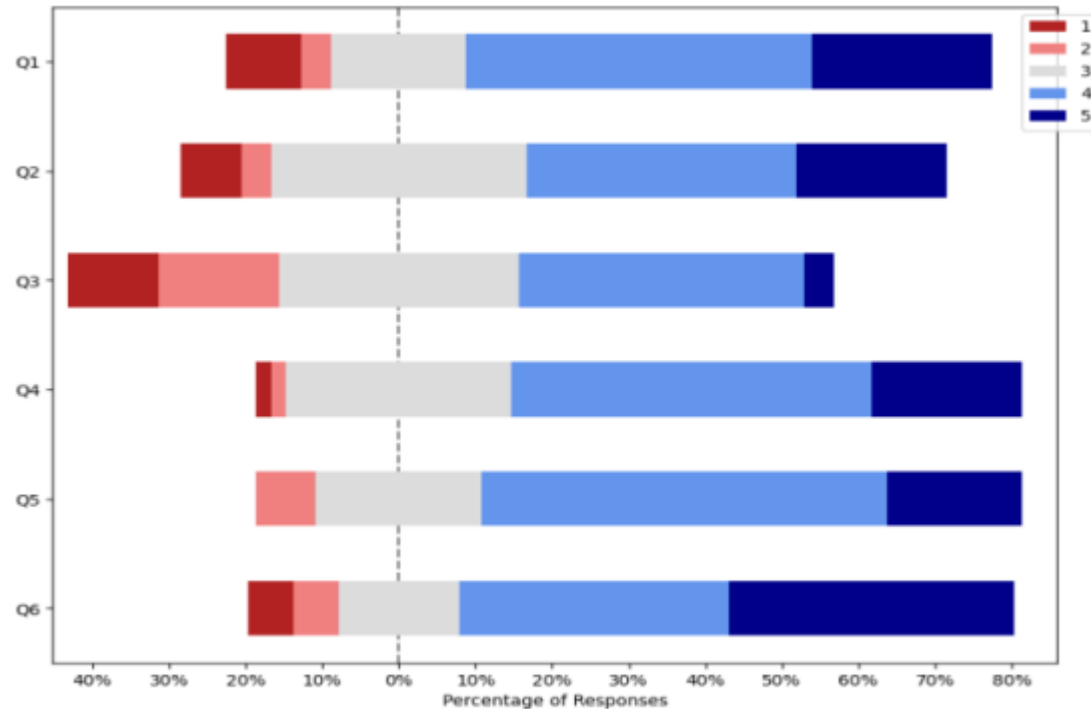


Figure 3. Diverging stacked bar chart - Likert questions for RQ2

- Q1 How critical is interdisciplinary collaboration in RAI development? (1: Not critical - 5: Extremely critical)
- Q2 How often do you collaborate with non-tech experts (e.g., ethics, law) in AI projects? (1: Never - 5: Always)
- Q3 How effective is your organization's ethical software/AI development training? (1: Not effective - 5: Very effective)

Figure 3. Diverging stacked bar chart - Likert questions for RQ2

結果 – RQ3

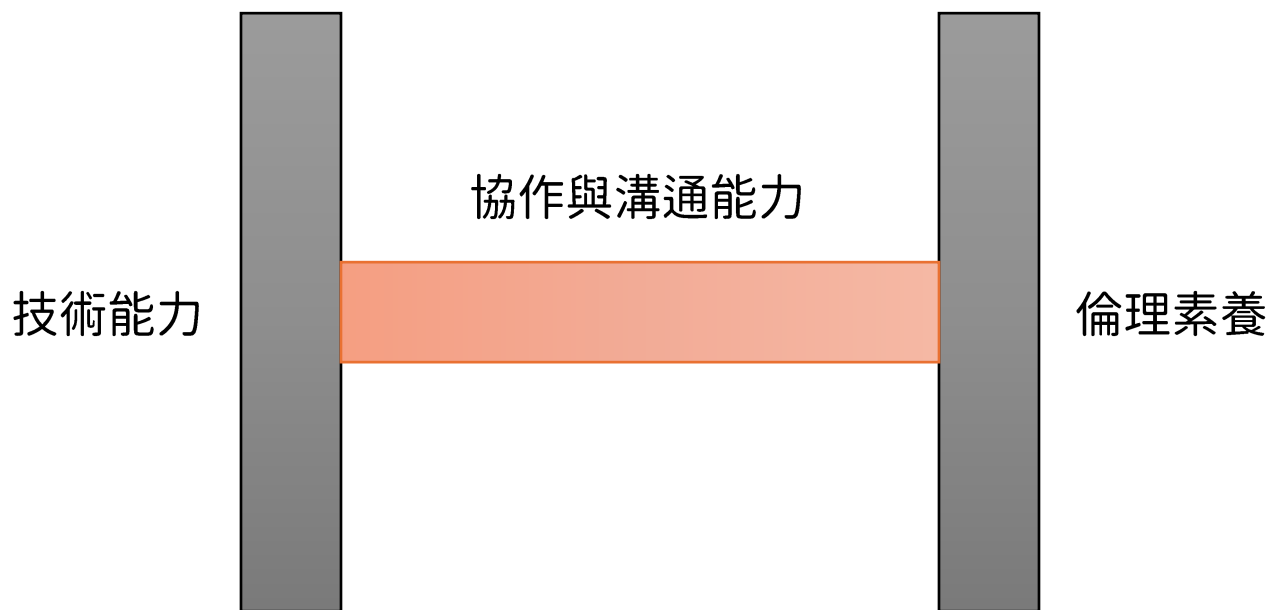


- Q1 How have your personal values influenced your approach to Software development or AI software development? (1: Not at all - 5: Extremely)
- Q2 To what extent did your values significantly influence the development of a responsible software/AI system? (1: Not at all - 5: Extremely)
- Q3 How much do conflicting personal values impact team decision-making in RAI development? (1: Negligible impact - 5: Substantial impact)
- Q4 To what extent do your personal values align with the organizational values? (1: Not at all - 5: Completely)
- Q5 How well do your ethical values in AI/software development align with your organization's ethical guidelines? (1: Not at all - 5: Completely)
- Q6 To what extent do you feel responsible for the ethical implications of the AI/Software systems you develop? (1: Not at all - 5: Completely)

Figure 4. Diverging stacked bar chart - Likert questions for RQ3

結果 – H型人才

- 同時擁有專業技術以及識別偏見與風險的倫理素養



總結

- AI開發數據審查
- H型人才
- 倫理審查SOP化